

Lines

If l is a line that is not vertical, then it has a well-defined slope.

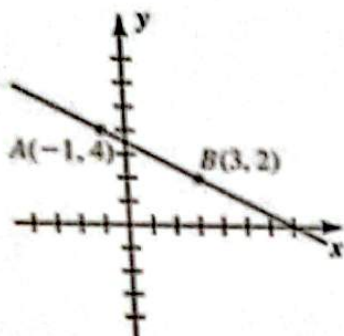
If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are any two distinct points on l , l has slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$

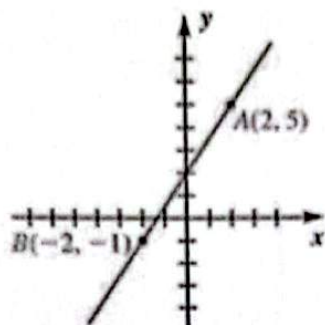
$$\neq \frac{y_1 - y_2}{x_2 - x_1}$$

Figure 3

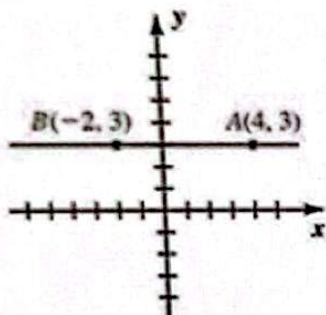
(a) $m = -\frac{1}{2}$



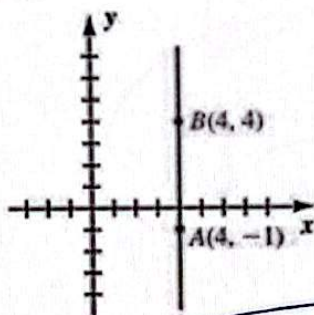
(b) $m = \frac{3}{2}$



(c) $m = 0$



(d) m undefined



(a) $m = \frac{2 - 4}{3 - (-1)} = \frac{-2}{4} = -\frac{1}{2}$

(b) $m = \frac{5 - (-1)}{2 - (-2)} = \frac{6}{4} = \frac{3}{2}$

(c) $m = \frac{3 - 3}{-2 - 4} = \frac{0}{-6} = 0$

(d) The slope is undefined because the line is parallel to the y-axis. Note that if the formula for m is used, the denominator is zero.

not from the same point

$$\frac{2 - 4}{-1 - 3} = \frac{-2}{-4} = \frac{1}{2} \times$$

Find an equation of the line through $A(-3, 4)$ that is parallel to

a) the x -axis

b) the y -axis



A diagram showing a line parallel to the x -axis. The line is labeled $x = -3$ and passes through the point $A(-3, 4)$. The line is drawn as a horizontal line segment with an arrow pointing to the right, and the equation $x = -3$ is written below it.



A diagram showing a line parallel to the y -axis. The line is labeled $y = 4$ and passes through the point $A(-3, 4)$. The line is drawn as a vertical line segment with an arrow pointing upwards, and the equation $y = 4$ is written to its left.

Point-Slope Form

If a line l has slope m and passes through $P_1(x_1, y_1)$, then it satisfies:

$$y - y_1 = m(x - x_1)$$

works for all lines that are not vertical

where $P(x, y)$ represents a general point on the line.

$$\div (x - x_1)$$

$$\frac{y - y_1}{x - x_1} = m$$

Q. Find an equation of a line through $A(1,7)$ and $B(-3,2)$.

$$y - y_1 = m(x - x_1)$$

needs $P_1(x_1, y_1)$ on the line

&
needs slope m

$$m = \frac{2 - 7}{-3 - 1} = \frac{-5}{-4} = \frac{5}{4}$$

Taking $P_1(x_1, y_1) = A(1, 7)$

$$x_1 = 1, y_1 = 7$$

$$y - 7 = \frac{5}{4}(x - 1)$$

If we take $P_1(x_1, y_1) = B(-3, 2)$

$$y - 2 = \frac{5}{4}(x - (-3))$$

$$y - 2 = \frac{5}{4}(x + 3)$$

$$y - 7 = \frac{5}{4}(x - 1)$$

Parallel & Perpendicular Lines

Two lines are parallel iff they have the same slopes

Two lines are perpendicular iff their slopes are negative reciprocals
(l_1 & l_2 are perp. iff $m_1 m_2 = -1$)

$$1) \quad \begin{aligned} y - 2 &= 3(x - (-5)) \\ y - (-1) &= -\frac{1}{3}(x - 0) \end{aligned}$$

$$m_1 = 3, \quad m_2 = -\frac{1}{3}$$

$$m_1 m_2 = 3 \cdot \left(-\frac{1}{3}\right) = -1$$

Find an equation for a line parallel to $6x + 3y = 4$ through $P(5, -7)$.

$$y - y_1 = m(x - x_1)$$

$$(x_1, y_1) \stackrel{?}{=} (5, -7)$$

$$x_1 = 5, y_1 = -7$$

$$y - (-7) = m(x - x_1)$$

$$y + 7 = m(x - 5)$$

If we can find slope of $6x + 3y = 4$, we know m .

We found $m = -2$

$$y + 7 = -2(x - 5)$$

d. Now do the same thing but perp. to $6x+3y=4$ and passing through $P(5, -7)$.

$$y - y_1 = m(x - x_1)$$

$$y - (-7) = m(x - 5)$$

$$y + 7 = m(x - 5)$$

We figured out
that this has slope (-2)
 m_1

What is m_2 so that

$$m_1 m_2 = -1$$

$$-2 m_2 = -1$$

$$m_2 = \frac{-1}{-2} = \frac{1}{2}$$

$\frac{-1}{-2}$
new slope

$$y + 7 = \frac{1}{2}(x - 5)$$